ALFIDO TECH – DATA ANALYTICS INTERNSHIP

TASK 3: EVENT ANALYSIS

Submitted by: Mirsaji S | **Tools Used:** Python, Pandas, NumPy, Matplotlib, Seaborn | **Environment:** Google Colab

PAGE 1: PROJECT OVERVIEW

1. Objective

The objective of this project is to analyze event data to understand event categories, websites, countries, artists, and daily event activity. The analysis helps identify common event patterns and understand how recorded events are distributed across different dimensions.

2. Dataset Overview

The event dataset contains event records with key dimensions including event categories (such as pageview, click, preview), associated websites, geographic country details (e.g., United States, India, France), artist attributes, and date timestamps tracking activity over time.

3. Executive Summary

This project analyzes event data using Python and data visualization techniques. The analysis focuses on event categories, the top 10 websites by event count, the top 10 countries by event count, the top 10 artists by event count, and daily event trends.

The reported results show that 'pageview' is the leading event category (>70,000 events), United States is the top country (28,664 events) followed by India (18,689 events), Tesher is the leading artist, and daily activity exhibited a peak on August 19, 2021 before stabilizing.

PAGE 2: EVENT ANALYSIS AND VISUALIZATIONS

4. Top Event Categories

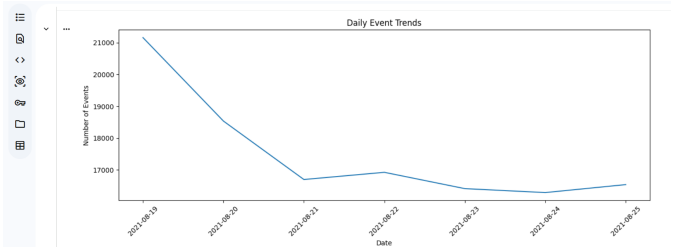


Figure 1: Top Event Categories

This analysis compares event counts across categories. The results show that 'pageview' recorded the highest count (>73,000), followed by 'click' (~32,500) and 'preview' (~16,800).

5. Top 10 Websites by Event Count

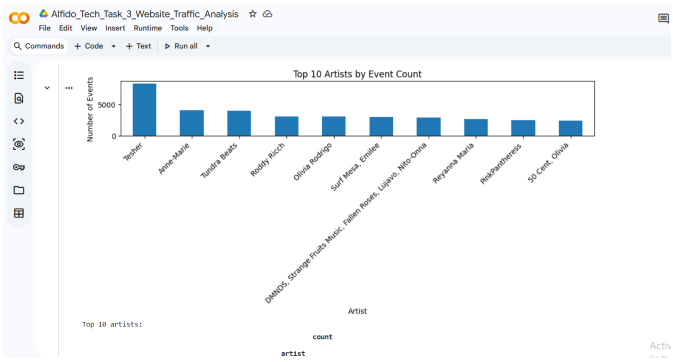


Figure 2: Top 10 Websites by Event Count

Website-wise analysis identifies the distribution of recorded events across web platforms, with 'pageview' driving the highest volume of events.

PAGE 3: COUNTRY, ARTIST AND DAILY EVENT ANALYSIS

6. Top 10 Countries by Event Count

Top 10 countries:

country	count
United States	28664
India	18689
France	10565
Saudi Arabia	7682
United Kingdom	5095
Germany	4017
Canada	2784
Pakistan	2633
Iraq	2444
Turkey	2399

Figure 3: Top 10 Countries by Event Count

Geographical analysis shows that the United States generated the highest event volume (28,664), followed by India (18,689), France (10,565), and Saudi Arabia (7,682).

7. Top 10 Artists by Event Count

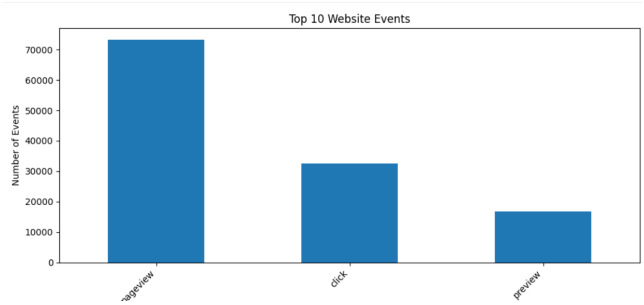


Figure 4: Top 10 Artists by Event Count

Artist analysis reveals that Tesher lead in total event counts (>7,500), followed by Anne-Marie, Tundra Beats, and Roddy Ricch.

8. Daily Event Count



Figure 5: Daily Event Count

Daily trend analysis shows peak event activity on August 19, 2021 (~21,100 events), followed by a gradual decrease to around 16,500 events per day.

PAGE 4: KEY FINDINGS AND CONCLUSION

9. Key Findings

1. **Leading Category:** 'pageview' is the most frequent event category (>73,000 events).
2. **Website Activity:** Events are strongly concentrated around pageviews and user clicks.
3. **Top Country:** United States leads globally with 28,664 events, followed by India with 18,689 events.
4. **Top Artist:** Tesher accumulated the highest event engagement among artists.
5. **Daily Activity:** Event activity peaked on August 19, 2021 before stabilizing between 16,000 and 17,000 daily events.

10. Recommendations

1. **Analyze Popular Categories:** Study the most common event categories to understand event distribution and optimize user engagement.
2. **Review Website Activity:** Examine websites with high event counts to understand where recorded events are concentrated.
3. **Compare Countries:** Use country-wise event counts to understand geographical patterns and tailor regional content.
4. **Study Artist Activity:** Analyze leading artists to identify music/creator preference trends.
5. **Monitor Daily Trends:** Review daily event counts to identify noticeable changes in event activity and operational peaks.

11. Conclusion

Python-based data analysis and visualization help summarize event data and identify patterns across categories, websites, countries, artists, and dates. These findings support a clearer understanding of overall event activity and user engagement.

12. Notebook Link

Task 3 – Event Analysis Notebook:
https://github.com/mirsaji01-star/Alfido_Tech_Data_Analytics/blob/main/Task_3_Website_Traffic_Analysis/Task_3_Notebook.ipynb

Internship Organization: Alfido Tech | Domain: Data Analytics | End of Report